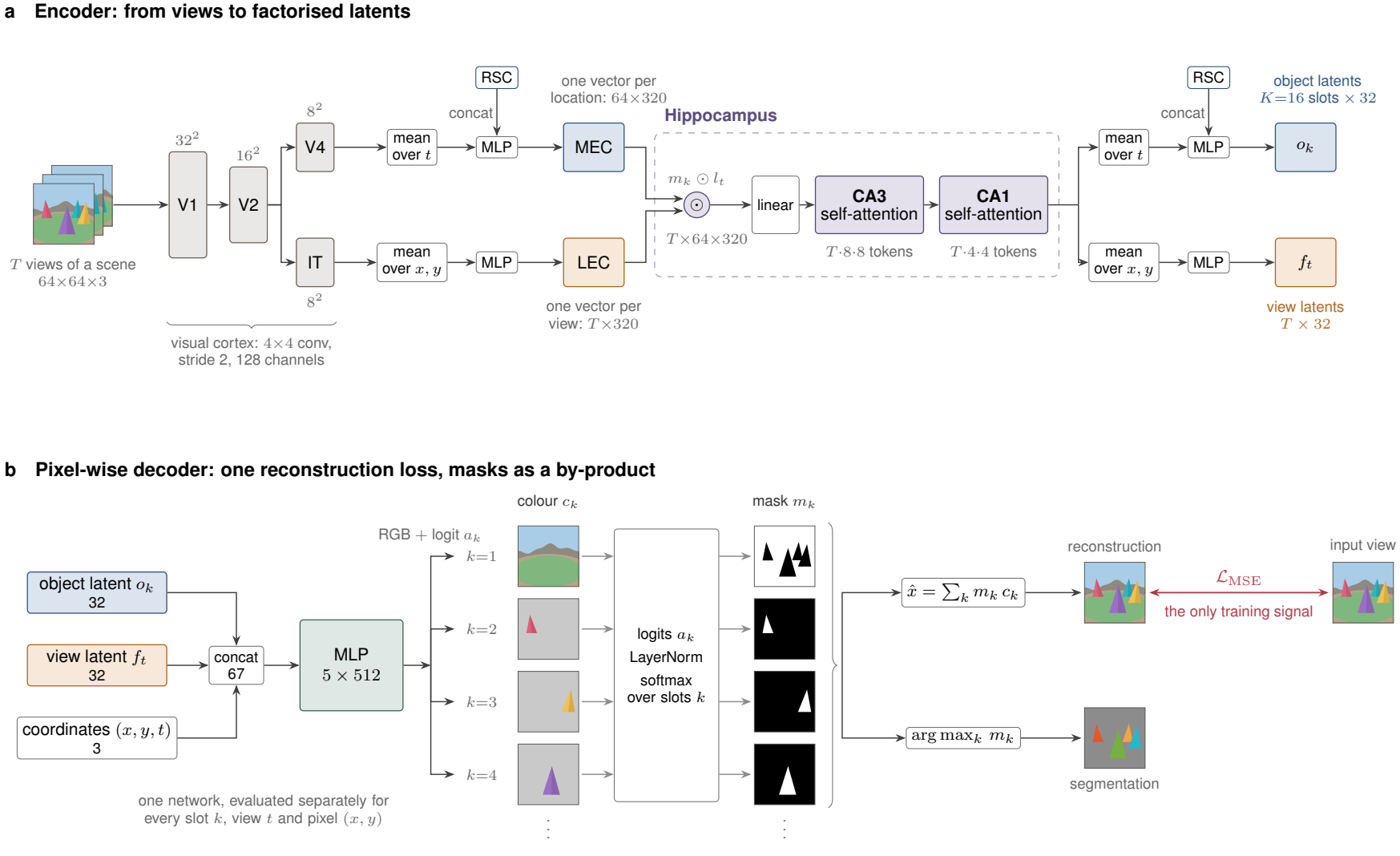

a Encoder: from views to factorised latents
T views of a scene
64×64×3
V1
V2
V4
IT
visual cortex: 4×4 conv, stride 2, 128 channels
mean over t
mean over x, y
MLP
RSC
concat
one vector per location: 64×320
MEC
LEC
one vector per view: T×320
Hippocampus
linear
CA3 self-attention
T·8·8 tokens
CA1 self-attention
T·4·4 tokens
T×64×320
object latents
K=16 slots × 32
view latents
T × 32
b Pixel-wise decoder: one reconstruction loss, masks as a by-product
object latent 32
view latent 32
coordinates (x, y, t) 3
concat 67
MLP 5 × 512
RGB + logit
colour
mask
logits
LayerNorm
softmax over slots k
reconstruction
input view
the only training signal
segmentation
one network, evaluated separately for every slot k, view t and pixel (x, y)